

Thread - 2024-06-05

<https://x.com/RiikonenMarko/status/1798387995524899163>

1/1 - 2024-06-05 16:14 UTC

Just putting this here for my reference. "120° rotated sup-uppercave Parry". The above sun part should be possible, actually I am a bit surprised that in Ounasvaara displays we have not yet seen this (3rd image shows low sun situation). The below sun part is made by the same crystals as the above part, but rotated 180 degrees. May be that such orientation is not realistic, that tabulars, just like (semi)triangulars, fall with their blunt end down. This halo is also made by semitriangulars, but less well.

HALOPOINT 2.0

File Tools Analysis Info

HALOPOINT 2.0

☒ Save work-file
☐ Save light work-file
☒ Open in Analysis Window

Start Simulation

Add to the Simulation

General Parameters

Sun Elevation [°]

Sun Diameter

Number of rays

Colour of Light:

☐ Single Wavelength [nm]

☒ Wavelength Range

From nm to nm

Number of hits allowed:

Ice Crystal Population Parameters

%	C - axis	Rotation	Prism Aspect Ratio	Upper Pyramid	Lower Pyramid
	Mean Std Distribution	Mean Std Distribution	Mean Std Distribution	Apex h Std Distribution	Apex h Std Distribution
☒ 1	4 90 0.3 Gaussian	90 0.3 Uniform	1 0.2 Uniform	56.14 0 0 Uniform	56.14 0 0 Uniform
☐ 2	4 90 0 Gaussian	90 0 Uniform	1 0 Uniform	56.142 0 0 Uniform	56.142 0 0 Uniform
☐ 3	4 90 0 Gaussian	90 0 Uniform	1 0 Uniform	56.142 0 0 Uniform	56.142 0 0 Uniform
☐ 4	20 90 0 Gaussian	90 360 Uniform	1 0 Uniform	56.142 0 0 Uniform	56.142 0 0 Uniform
☐ 5	20 90 0 Gaussian	90 360 Uniform	1 0 Uniform	56.142 0 0 Uniform	56.142 0 0 Uniform
☐ 6	20 90 0 Gaussian	90 360 Uniform	1 0 Uniform	56.142 0 0 Uniform	56.142 0 0 Gaussian
☒ 7	4 90 0.3 Gaussian	-90 0.3 Uniform	1 0.2 Uniform	56.142 0 0 Gaussian	56.142 0 0 Gaussian
☐ 8	26 90 0.1 Gaussian	0 0.1 Uniform	0.5 0 Uniform	165 0 0 Uniform	165 0 0 Uniform
☐ 9	1 0 0.2 Gaussian	30 0.2 Uniform	1 0.00 Uniform	56.142 0 0 Uniform	56.142 0 0 Uniform
☐ 10	1 90 0.2 Gaussian	90 0.2 Gaussian	4 0.1 Uniform	56.142 0 0 Gaussian	56.142 0 0 Gaussian

Simulation View

Lens Type

Focal Length [mm]

Sensor Size [mm] W: H:

☐ Custom parameters a: b:

Center of Image

Azimuth:

Elevation:

☒ Portrait ☐ Landscape Rotation (CW)

Ice Crystal Visualisation

View Angle Control

y

z

Line Width, Front

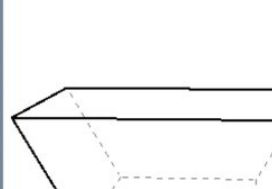
Line Width, Back

Dash Length (Back)

Dash Spacing (Back)

Back Line Colour

Ice Crystal Population



Prism Face Distances

Apply	Face Distance from c-axis	Std of Distance	Sync	Distribution
	3 4 5 6 7 8	3 4 5 6 7 8		
☒ 1	1 2 1 0.3 1 2	0 0 0 1 0 0	☐	Uniform
☒ 2	1 2 1 0.3 1 2	0 0 0 1 0 0	☐	Uniform
☒ 3	1 2 1 0.3 1 2	0 0 0 1 0 0	☐	Uniform
☐ 4	1 2 1 0.5 1 2	0 0 0 1 0 0	☐	Uniform
☐ 5	1.7 1 1.7 1 1.7 1	0.2 0 0.2 0 0.2 0	☒	Uniform
☐ 6	1 2 1 2 1 2	0 0 0 0 0 0	☐	Uniform
☒ 7	1 2 1 0.3 1 2	0 0 0 1 0 0	☐	Uniform
☐ 8	2 1 2 1 2 1	0 0 0 0 0 0	☐	Uniform
☒ 9	1 1 1 1 1 1	0.3 0.3 0.3 0.3 0.3 0.3	☐	Uniform
☐ 10	1 1 1 1 1 1	0 0 0 0 0 0	☐	Gaussian

Appearance

Dots

Background

Shade Levels

☐ Show Coordinate Grid

Major Step [°]

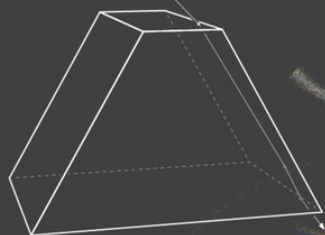
Minor Step [°]

Style

☒ Horizon ☐ Hide Subhorizon

"120° rotated sub-uppercave Parry"

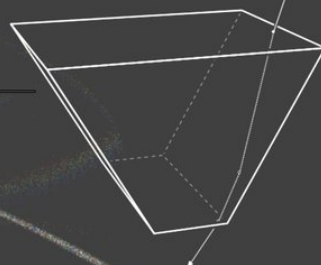
346



above sun part

below sun part

356



120°

